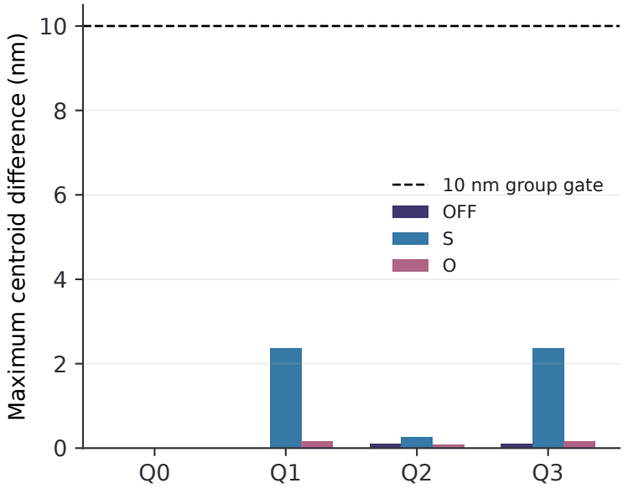
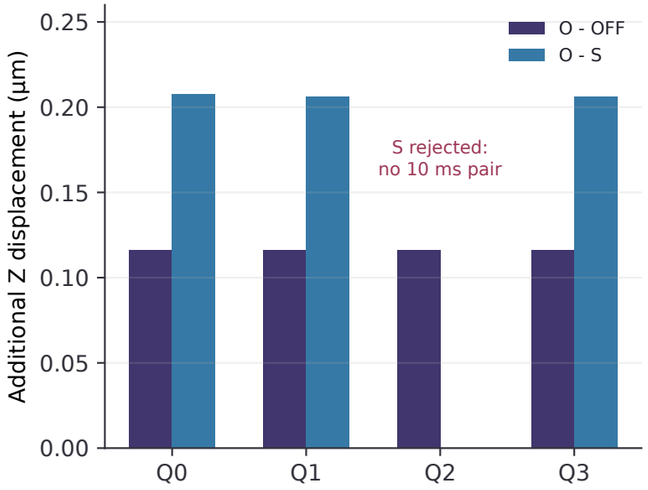
F10C | External-query caching preserves centroid motion better than morphology

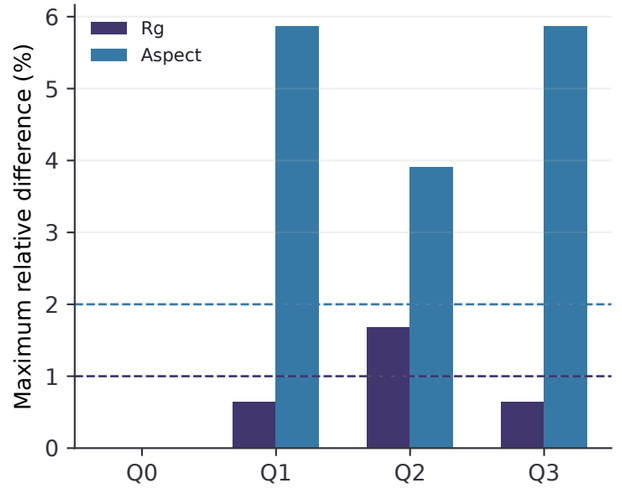
A Group motion vs Q0



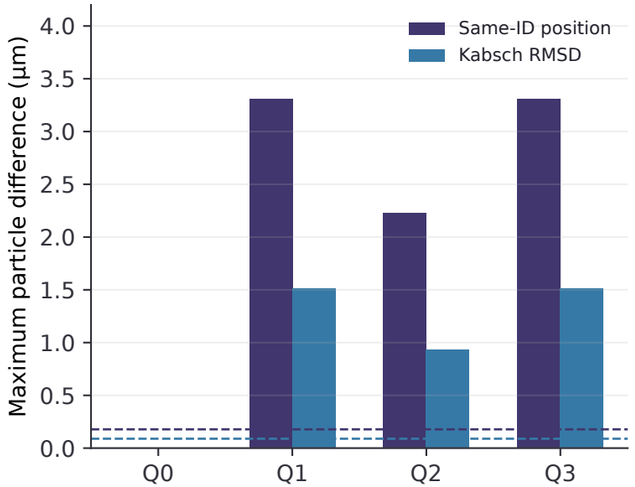
B Matched 10 ms endpoints



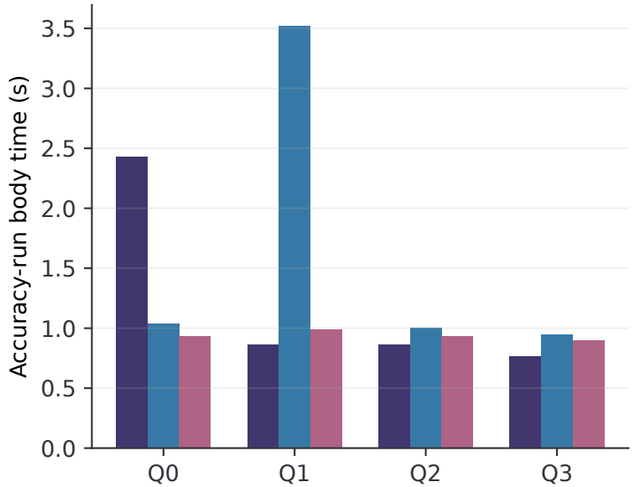
C Shape metrics and registered gates



D Internal rearrangement differs



E Recorded costs; no admitted speed claim



F Separate numerical admission

Mode	Group	Morph.	Full model
Q0	PASS	PASS	BASELINE
Q1	PASS	FAIL	REJECT
Q2	FAIL	FAIL	REJECT
Q3	PASS	FAIL	REJECT

Q2-S: first rejection at 9.16 ms
Overlap 17.865 nm > 17.8 nm budget
No q = 4 or batch + cache production run

Q0: (acoustic, flow) stride (1,1); Q1: (2,1); Q2: (1,2); Q3: (2,2). Mechanics and CFD proposal validity checked every 10 us.
Q2-S error bars use its accepted prefix through 9.15 ms. Exact edge-event timing is required for morphology; all ON-cache modes fail this gate.